

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

Shonika (192511387)

I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1. Network Throughput and Packet Loss

START

INPUT total_bits = 8000

INPUT time = 4

INPUT packets_sent = 200

INPUT packets_received = 180

IF time = 0 OR packets_sent = 0 THEN

 PRINT "Invalid input"

 STOP

END IF

throughput = total_bits / time

packet_loss = ((packets_sent - packets_received) / packets_sent) * 100

PRINT "Throughput =", throughput, "bps"

PRINT "Packet Loss =", packet_loss, "%"

IF throughput > 1500 AND packet_loss < 10 THEN

 PRINT "Network Status: Efficient"

ELSE

 PRINT "Network Status: Inefficient"

END IF

PRINT "Network Performance Summary"

STOP

Result:

Throughput = **2000 bps**

Packet Loss = **10%**

Network Status = **Inefficient**

(Because packet loss is not less than 10%.)

2. Bandwidth Utilization Monitoring

START

INPUT number_of_links

CREATE usage[]

WHILE TRUE

FOR i = 1 TO number_of_links

INPUT bandwidth_usage

usage[i] = bandwidth_usage

IF bandwidth_usage > 80% THEN

PRINT "Link", i, "is overloaded"

PRINT "Switch traffic to backup link"

ELSE

PRINT "Link", i, "is normal"

END IF

END FOR

WAIT for next monitoring interval

END WHILE

STOP

Explanation:

The algorithm continuously monitors all links and stores bandwidth usage in an array. If utilization exceeds **80%**, traffic is shifted to a backup link to reduce congestion.

3. Best Communication Path Selection

START

INPUT number_of_links

CREATE bandwidth[], delay[], utilization[], score[]

WHILE TRUE

FOR each link

INPUT bandwidth

INPUT delay

INPUT utilization

IF link is failed OR utilization > 80% THEN

PRINT "Link overloaded/failed"

SELECT alternate path

ALERT administrator

ELSE

score = bandwidth - delay - utilization

END IF

END FOR

SELECT path with highest score

PRINT "Best Path Selected"

WAIT for next monitoring interval

END WHILE

STOP

Explanation:

The algorithm continuously monitors bandwidth, delay, and utilization, calculates a link-quality score, and selects the path with the highest score. If a link fails or becomes overloaded, an alternate path is selected automatically and the administrator is alerted. This improves reliability, routing efficiency, and fault tolerance.